

CLIMATE MASTERY

Why Hardwood Succeeds or Fails in Toronto

The physical science, moisture protocol, and non-negotiable steps that turn a floor into a permanent asset — or a permanent liability

Hardwood, Done Once. Done Right.

ecowoods.ca | Est. 1998 / 2000 | 5,200+ Homes | Lifetime Workmanship Warranty

The Physical Reality Stack

- 1 Toronto's climate is extreme for wood — not mild
- 2 Wood is hygroscopic and anisotropic
- 3 Moisture testing is the first non-negotiable gate
- 4 Solid versus engineered — what the house can actually support
- 5 Installation method must be dictated by the substrate
- 6 The complete EcoWoods Non-Negotiable Protocol
- 7 Visible failure modes when any step is skipped
- 8 How lifetime workmanship becomes possible (and defensible)

Toronto's Climate Reality

- Winter indoor RH routinely crashes to **18–25%**
- Summer indoor RH regularly exceeds **60%**
- Ideal band for hardwood: **35–55% RH**
- This annual swing is among the most aggressive of any major North American residential market

Core Truth

Wood will move. The only question is whether the floor was specified, tested, acclimated, and installed to survive decades of that movement.

Key Numbers

- Winter low: 18–25% RH
- Summer high: >60% RH
- Safe operating band: 35–55% RH
- Primary movement: across the grain

- Hardwood constantly exchanges moisture with the surrounding air
- Expansion and contraction occur primarily **across the grain**
- Length change is minimal; width change is significant and cumulative
- Every board arrives with its own moisture content history

the floor will eventually declare the error — permanently. ;

Moisture Testing — The First Non-Negotiable

- 1 Subfloor moisture content must be measured
- 2 Flooring material moisture content must be measured
- 3 Both readings must sit within the manufacturer's and EcoWoods' acceptable delta
- 4 Testing occurs on the free estimate **and** again immediately before installation

Why this is non-negotiable

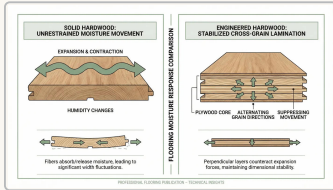
A floor installed over a wet subfloor or with boards that have not equalized will cup, crown, gap, or buckle. There is no warranty language that can override physics.



Moisture testing on site — non-negotiable

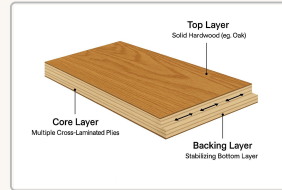
Solid vs Engineered in the GTA

Solid Hardwood



- Typically 3/4 (19 mm)
- Generational wear layer
- Highly sensitive to RH swings
- Best over plywood with nail-down

Engineered Hardwood



- Real hardwood wear layer + 90° cross-ply core
- Dimensional stability
- Survives concrete, condos, radiant heat

EcoWoods Position: Engineered is the correct specification for the majority of Toronto projects.
We specify what the house can support. We never sell what will fail.

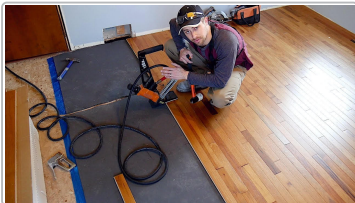
Correct Method Matched to Substrate

- 1 **Nail-down** → solid hardwood over plywood
- 2 **Glue-down** → engineered over concrete or in condominiums
- 3 **Floating** → engineered over radiant or when acoustic separation is required

Critical Discipline

The method is not a preference or a sales option.

It is determined by the substrate, the product construction, and the climate load the floor will face for decades.



The EcoWoods Non-Negotiable Protocol

- 1 Moisture testing of subfloor + material (documented)
- 2 Minimum 72-hour acclimation in the actual conditioned space
- 3 Correct installation method matched to substrate
- 4 Proper expansion gaps at all fixed objects and walls
- 5 99.7% HEPA dust containment throughout the process

Dual Purpose

These steps prevent physical failure **and** create the psychological authority that allows EcoWoods to offer a lifetime workmanship warranty and still sleep at night.

When Any Step Is Skipped

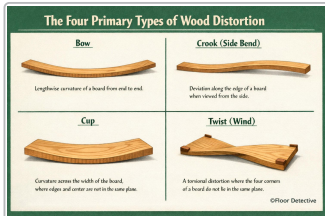
Visible Failures

- Cupping (edges higher than centre)
- Crowning (centre higher than edges)
- Seasonal gapping
- Buckling / tenting
- Edge peaking and finish failure

The Sentence

“The floor will declare your incompetence.”

These are not aesthetic issues.
They are permanent, visible records of process failure
that no amount of later refinishing can fully erase.



What the Intelligent Homeowner Must Demand

- 1 Written moisture readings of both subfloor and material before any deposit
- 2 Explicit confirmation of minimum 72-hour acclimation in the actual space
- 3 Fixed price in writing with no open-ended “unforeseen conditions” language
- 4 Confirmation that only salaried master artisans will perform the work
- 5 Lifetime workmanship warranty in the contract
- 6 Willingness of the company to refuse the job if conditions are wrong

Any company that cannot or will not provide these is optimizing for speed and lowest bid, not for decades of performance.

How Lifetime Workmanship Becomes Possible

- Only salaried master artisans — never subcontractors
- Fixed price in writing — the number never moves
- Every step of the protocol is documented and retained
- The same ruthless standard applied to every square foot
- Willingness to refuse the job that physics says will fail

Physical mastery is the only foundation
on which a lifetime warranty can rest.

Key Takeaways for Homeowners

- 1 Toronto's climate is hostile to poorly installed hardwood
- 2 Moisture testing and proper acclimation are not optional extras
- 3 Engineered is usually the correct choice for condos, concrete, and radiant
- 4 Method must match substrate — no exceptions, no preferences
- 5 Demand the full protocol in writing before any deposit is paid
- 6 The company that refuses the job that will fail is the company you want

The floor will declare your competence.

Make sure it only ever declares mastery.

EcoWoods

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